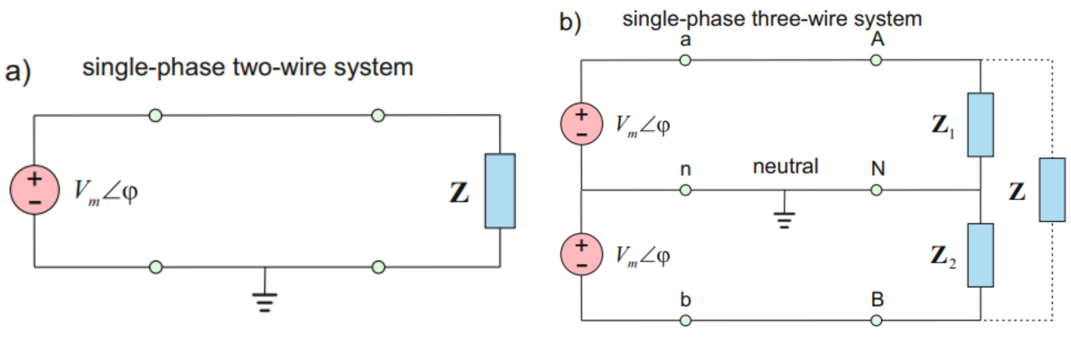
Contents

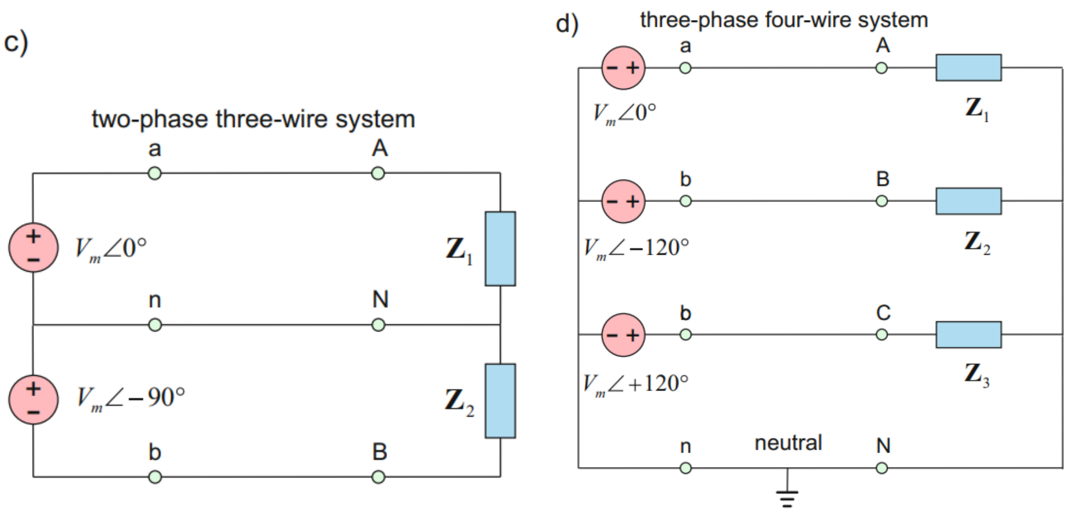
Chapter 1

Balanced Three phase circuit

1.1 Single phase vs Polyphase circuits



1. (Left) Single phase ac power system (단상2선식) consists of three parts
   - Generator : AC source $V_p\angle\phi$, V_p is the rms magnitude of the source voltage
   - Transmission line represented as a pair or wires
   - Load represented by an impedance $\mathbf{Z_L}$
2. (Right) A common household system employs a single phase three-wire system (단상3선식), which consists of two identical sources connected by two loads through two outer wires and a neutral wire. In this system,
   - The terminal voltages have the same magnitude and the same phase.
   - both 120-V ($\mathbf{Z_1}$ at A-N terminal and $\mathbf{Z_2}$ at N-B terminal) and 240-V ($\mathbf{Z}$ at A-B terminal) appliances can be connected.
3. Now, we are going to deal with circuits or systems in which the ac sources operate at the same frequency but **different phases**, so-called **polyphase** systems.



4. (Left) Two-phase three-wire system (2상 3선식) is produced by a generator consisting of two coils **perpendicular** to each other so that the voltage generated by one lags the other by 90°.

5. (Right) Three-phase four-wire system (3상 4선식) is produced by a generator consisting of three sources having the same amplitude and frequency but **out of phase with each other by** 120°. This system is by far the most prevalent and most economical polyphase system.

6. Three phase systems are important for at least three reasons:

   - Nearly all electric power is generated and distributed in three-phase at the operating frequency of 60 Hz (or $\omega = 377$ rad/s) in the United States or 50 Hz (or $\omega = 314$ rad/s) in some other parts of the world.
     When one-phase or two-phase inputs are required, they are taken from the three-phase system rather than generated independently. Even when more than three phases are needed—such as in the aluminum industry, where 48 phases are required for melting purposes—they can be provided by manipulating the three phases supplied.
   - the instantaneous power in a three-phase system can be constant (not pulsating), resulting in uniform power transmission and less vibration of three-phase machines.
   - For the same amount of power, the three-phase system is more economical than the single-phase. The amount of wire required for a three-phase system is less than that required for an equivalent single-phase system.

1.2 Balanced Three-Phase Voltages

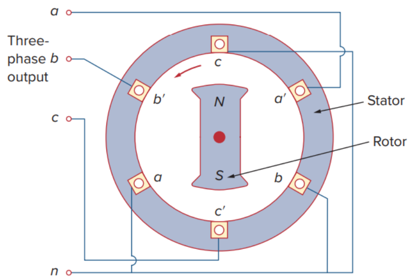
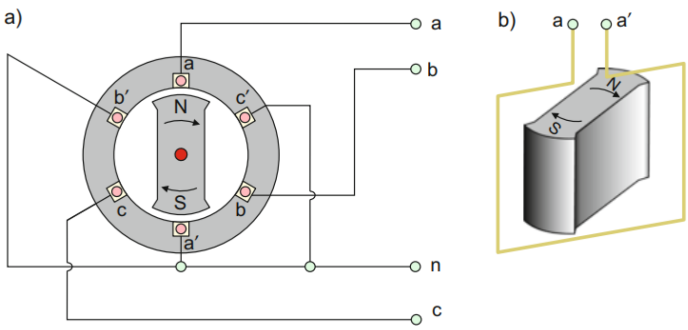
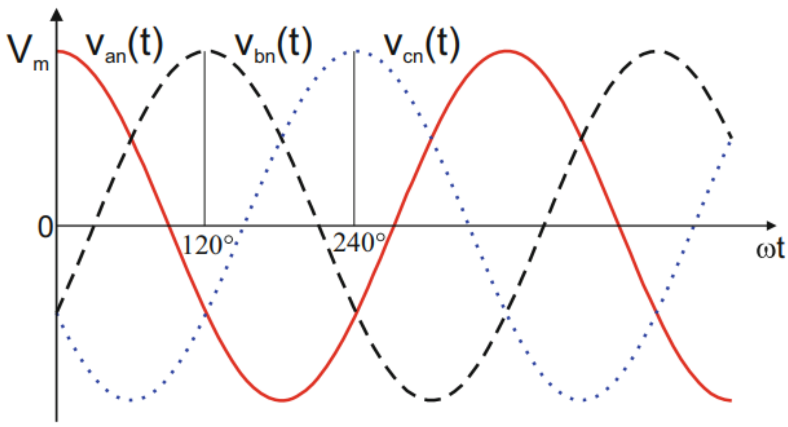


Figure 12.4
A three-phase generator.



1. Three phase voltages are often produced with a three-phase ac generator whose cross-sectional view is shown in Fig. 12.4.
   - Three-phase ac generator = rotor(rotating magnet) + stator (stationary winding)
   - Three separate windings or coils with terminals a-a′, b-b′, and c-c′ are physically placed 120° apart around the stator. Terminals a and a′, for example, stand for one of the ends of coils going into and the other end coming out of the page. As the rotor rotates, its magnetic field "cuts" the flux from the three coils and induces voltages in the coils. Because the coils are placed 120° apart, the induced voltages in the coils are equal in agnitude but out of phase by 120°
   - Since each coil can be regarded as a single-phase generator by itself, the three-phase generator can supply power to both single-phase and three-phase loads.
2. The example of three phase voltage produced by three-phase ac generator is given below:



3. A typical three-phase system consists of three **voltage** sources connected to loads by three or four wires (or transmission lines). (Three phase current sources are very scarce.)
4. A three-phase system is **equivalent** to three single-phase circuits.

5. The voltage sources can be either wye-connected as shown in Fig. 12.6(a) or delta-connected as in Fig. 12.6(b).

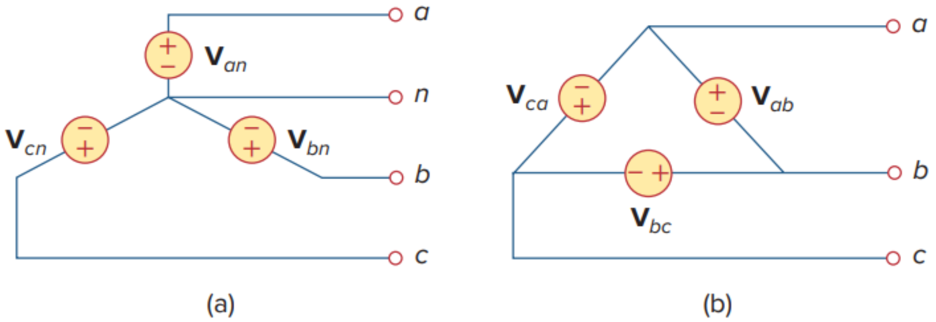


Figure 12.6
Three-phase voltage sources: (a) Y-connected source, (b) Δ-connected source.

6. Let us consider the **wye**-connected voltages in Fig. 12.6(a) for now. The voltages $\mathbf{V}_{an}$, $\mathbf{V}_{bn}$, and $\mathbf{V}_{cn}$ are respectively between lines a, b, and c, and the neutral line n. These voltages are called **phase voltages**.

7. **Balanced** phase voltages are equal in magnitude and are out of phase with each other by 120°, satisfying

$$\mathbf{V}_{an} + \mathbf{V}_{bn} + \mathbf{V}_{cn} = 0$$
$$|\mathbf{V}_{an}| = |\mathbf{V}_{bn}| = |\mathbf{V}_{cn}|$$

This condition can be met in two sequences of phase voltages:

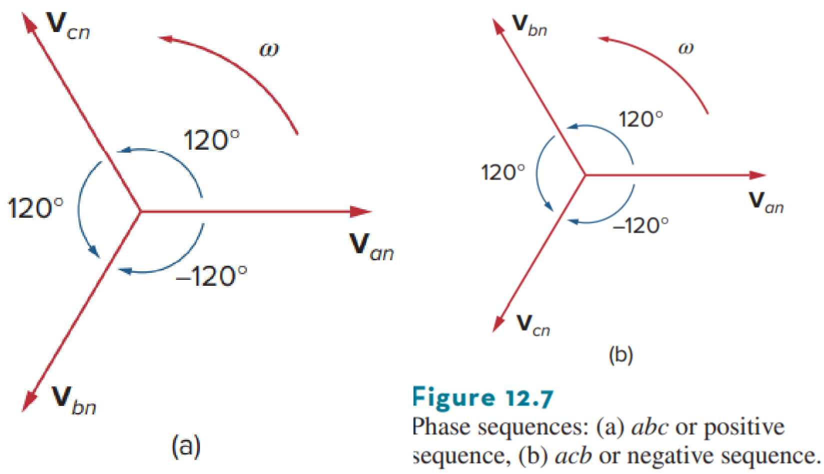


Figure 12.7
Phase sequences: (a) *abc* or positive sequence, (b) *acb* or negative sequence.

- Positive(abc) sequence : $\mathbf{V}_{an} = V_p\angle 0°, \mathbf{V}_{bn} = V_p\angle -120°, \mathbf{V}_{cn} = V_p\angle -240°$
- Negative(acb) sequence : $\mathbf{V}_{an} = V_p\angle 0°, \mathbf{V}_{cn} = V_p\angle -120°, \mathbf{V}_{bn} = V_p\angle -240°$

Remark. The phase sequence is the time order in which the voltages pass through their respective maximum values. This is determined by the order in which the phasors pass through a fixed point in the phase diagram.
Remark. The phase sequence is important in three-phase power distribution. It determines the direction of the rotation of a motor connected to the power source, for example.

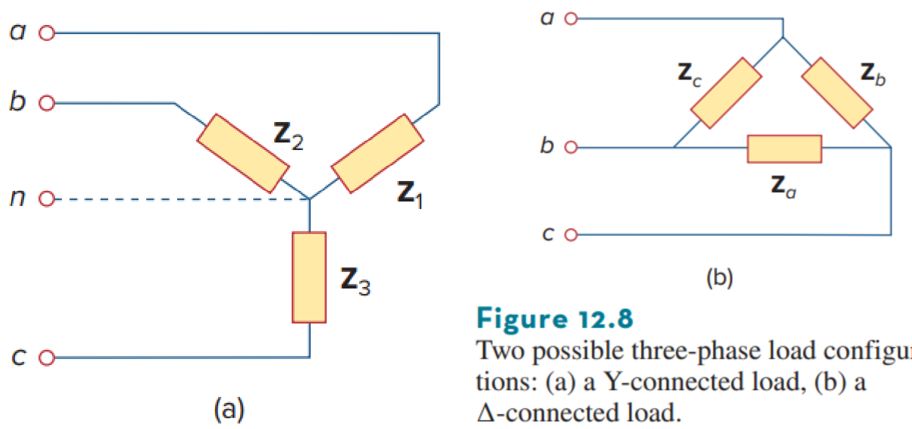


Figure 12.8
Two possible three-phase load configurations: (a) a Y-connected load, (b) a Δ-connected load.

8. Like the generator connections, a three-phase load can be either wye-connected or delta-connected, depending on the end application. Figure 12.8(a) shows a wye-connected load, and Fig. 12.8(b) shows a delta-connected load. The neutral line in Fig. 12.8(a) may or may not be there, depending on whether the system is four- or three-wire. (And, of course, a neutral connection is topologically impossible for a delta connection.)

9. A **balanced** load is one in which the phase impedances are equal in magnitude and in phase. For a *balanced* wye-connected load,

$$\mathbf{Z}_1 = \mathbf{Z}_2 = \mathbf{Z}_3 = \mathbf{Z}_Y \tag{1.1}$$

where $\mathbf{Z}_Y$ is the load impedance per phase. For a *balanced* delta-connected load,

$$\mathbf{Z}_a = \mathbf{Z}_b = \mathbf{Z}_c = \mathbf{Z}_\Delta \tag{1.2}$$

where $\mathbf{Z}_\Delta$ is the load impedance per phase in this case. We recall from wye-delta transformation that

$$\mathbf{Z}_\Delta = 3\mathbf{Z}_Y \quad \text{or} \quad \mathbf{Z}_Y = \frac{1}{3}\mathbf{Z}_\Delta \tag{1.3}$$

10. Because both the three-phase source and the three-phase load can be either wye- or delta-connected, we have four possible connections:

    - Y-Y connection (i.e., Y-connected source with a Y-connected load).
    - Y-Δ connection.
    - Δ-Δ connection.
    - Δ-Y connection.

11. **Remark 1.** A balanced delta-connected load is more common than a balanced wye-connected load. This is due to the ease with which loads may be added or removed from each phase of a delta-connected load. This is very difficult with a wye-connected load because the neutral may not be accessible.
    Remark 2. Delta-connected sources are not common in practice because of the circulating current that will result in the delta-mesh if the three-phase voltages are slightly unbalanced.

1.3 Balanced Wye-Wye connection

1. **Motivation.** We begin with the Y-Y system, because **any** balanced three-phase system can be reduced to an equivalent Y-Y system. Therefore, analysis of this system should be regarded as the key to solving all balanced three-phase systems.

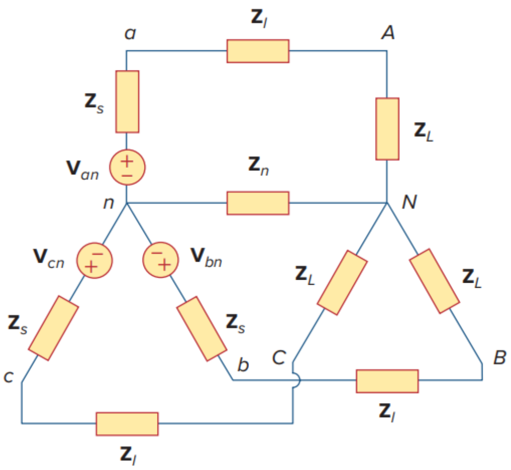


Figure 12.9
A balanced Y-Y system, showing the source, line, and load impedances.

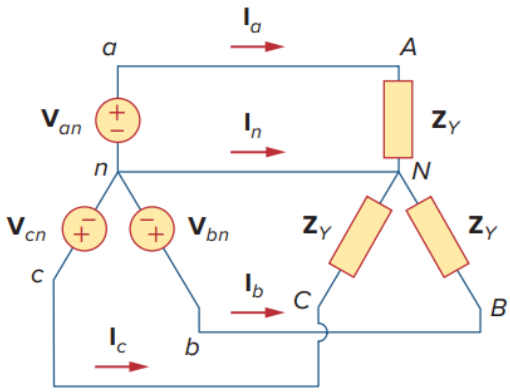


Figure 12.10
Balanced Y-Y connection.

2. A **balanced Y-Y system** is a three-phase system with a balanced Y-connected source and a balanced Y-connected load.

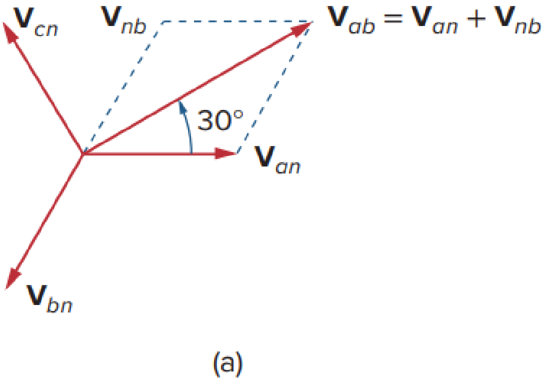
3. $\mathbf{Z}_Y$ is the total load impedance **per phase** and is the sum of the sum of the source impedance $\mathbf{Z}_s$, line impedance $\mathbf{Z}_l$, and load impedance $\mathbf{Z}_L$ for each phase, since these impedances are in series.

$$\mathbf{Z}_Y = \mathbf{Z}_s + \mathbf{Z}_l + \mathbf{Z}_L$$

   where

   - $\mathbf{Z}_s$: the internal impedance of the phase winding of the generator
   - $\mathbf{Z}_l$: the impedance of the line joining a phase of the source with a phase of the load (transmission line impedance)
   - $\mathbf{Z}_L$: the impedance of each phase of load

4. $\mathbf{Z}_s$ and $\mathbf{Z}_l$ are often very small compared with Z_L, so one can assume that $\mathbf{Z}_Y = \mathbf{Z}_L$ if no source or line impedance is given. In any event, by **lumping** the impedances together, the Y-Y system in Fig. 12.9 can be simplified to that shown in Fig. 12.10.



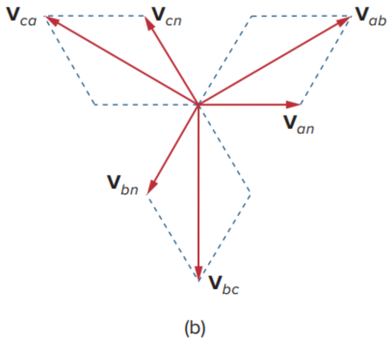


Figure 12.11
Phasor diagrams illustrating the relationship between line voltages and phase voltages.

5. **Phase voltages vs Line voltages.**
   Assuming the positive(abc) sequence, the phase voltages are

$$\mathbf{V}_{an} = V_p\angle 0°, \mathbf{V}_{bn} = V_p\angle -120°, \mathbf{V}_{cn} = V_p\angle -240° = V_p\angle +120°$$

   The line-to-line voltages or simply **line** voltages $\mathbf{V}_{ab}$, $\mathbf{V}_{bc}$, and $\mathbf{V}_{ca}$ are related to the phase voltages as follows:

$$\begin{aligned}\mathbf{V}_{ab} &= \mathbf{V}_{an} + \mathbf{V}_{nb} \\ &= \mathbf{V}_{an} - \mathbf{V}_{bn} \\ &= V_p\angle 0° - V_p\angle -120° = V_p\angle 0° + V_p\angle 60° \\ &= V_p\left[1 + \left(\frac{1}{2} + j\frac{\sqrt{3}}{2}\right)\right] = \sqrt{3}V_p\left[\frac{\sqrt{3}}{2} + j\frac{1}{2}\right] \\ &= \sqrt{3}V_p\angle 30°\end{aligned}$$

   Similarly,

$$\begin{aligned}\mathbf{V}_{bc} &= \sqrt{3}V_p\angle -90° \\ \mathbf{V}_{ca} &= \sqrt{3}V_p\angle -210°\end{aligned}$$

   Figure 12.11(a) illustrates this. Figure 12.11(a) also shows how to determine the line voltage $\mathbf{V}_{ab}$ from the phase voltages, while Fig. 12.11(b) shows the same for the three line voltages.

6. To summarize, the magnitude of the line voltages V_L is $\sqrt{3}$ times the magnitude of the phase voltages V_p, or

$$V_L = \sqrt{3}V_p$$

   where

$$V_p = |\mathbf{V}_{an}| = |\mathbf{V}_{bn}| = |\mathbf{V}_{cn}|, \quad V_L = |\mathbf{V}_{ab}| = |\mathbf{V}_{bc}| = |\mathbf{V}_{ca}|$$

   Also the line voltages lead their corresponding phase voltages by 30°. Note that three line voltages also sum up to zero as do the phase voltages.

$$\mathbf{V}_{ab} + \mathbf{V}_{bc} + \mathbf{V}_{ca} = 0$$

7. **Line currents of the balanced Y-Y system.** Applying KVL to each phase in Fig. 12.10, we obtain the line currents as

$$\mathbf{I}_a = \frac{\mathbf{V}_{an}}{\mathbf{Z}_Y}, \quad \mathbf{I}_b = \frac{\mathbf{V}_{bn}}{\mathbf{Z}_Y} = \frac{\mathbf{V}_{an}/\angle -120^\circ}{\mathbf{Z}_Y} = \mathbf{I}_a/\angle -120^\circ$$

$$\mathbf{I}_c = \frac{\mathbf{V}_{cn}}{\mathbf{Z}_Y} = \frac{\mathbf{V}_{an}/\angle -240^\circ}{\mathbf{Z}_Y} = \mathbf{I}_a/\angle -240^\circ$$

We can readily infer that the line currents add up to zero,

$$\mathbf{I}_a + \mathbf{I}_b + \mathbf{I}_c = 0$$

so that

$$\mathbf{I}_n = -(\mathbf{I}_a + \mathbf{I}_b + \mathbf{I}_c) = 0 \tag{12.17a}$$

or

$$\mathbf{V}_{nN} = \mathbf{Z}_n \mathbf{I}_n = 0 \tag{12.17b}$$

that is, the voltage across the neutral wire is zero. The neutral line can thus be removed without affecting the system.
In fact, in long distance power transmission, conductors in multiples of three are used with the earth itself acting as the neutral conductor. Power systems designed in this way are well grounded at all critical points to ensure safety.

8. While the *line* current is the current in each line, the *phase* current is the current in each phase of the source or load. In the Y-Y system, the line current is the **same** as the phase current.

9. An alternative way of analyzing a balanced Y-Y system is to do so on a **"per phase" basis**. We look at one phase, say phase a, and analyze the single-phase

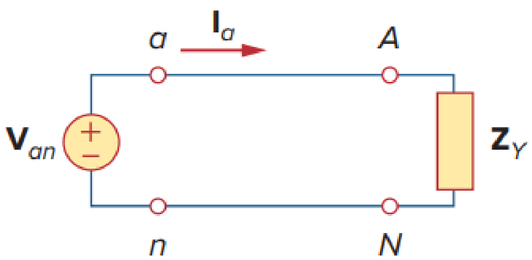


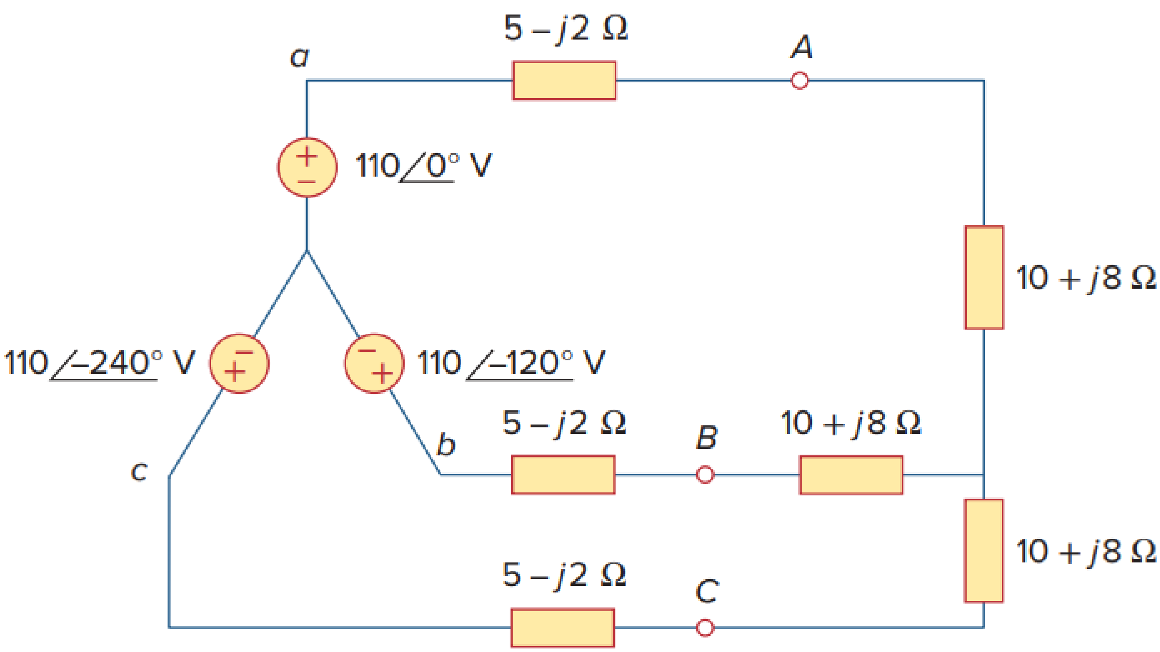
Figure 12.12
A single-phase equivalent circuit.

equivalent circuit in Fig. 12.12. The single-phase analysis yields the line current $\mathbf{I}_a$ as

$$\mathbf{I}_a = \frac{\mathbf{V}_{an}}{\mathbf{Z}_Y}$$

From $\mathbf{I}_a$, we use the phase sequence to obtain other line currents. **Thus, as long as the system is balanced, we need only analyze one phase**. We may do this even if the neutral line is absent, as in the three-wire system.

10. **Example:** Calculate the line currents in the three-wire Y-Y system of Fig. 12.13.



Solution:
The three-phase circuit in Fig. 12.13 is balanced; we may replace it with its single-phase equivalent circuit such as in Fig. 12.12. We obtain $\mathbf{I}_a$ from the single-phase analysis as

$$\mathbf{I}_a = \frac{\mathbf{V}_{an}}{\mathbf{Z}_Y}$$

where $\mathbf{Z}_Y = (5 - j2) + (10 + j8) = 15 + j6 = 16.155/\angle 21.8^\circ$. Hence,

$$\mathbf{I}_a = \frac{110/\angle 0^\circ}{16.155/\angle 21.8^\circ} = 6.81/\angle - 21.8^\circ \text{ A}$$

In as much as the source voltages in Fig. 12.13 are in positive sequence, the line currents are also in positive sequence:

$$\mathbf{I}_b = \mathbf{I}_a/\angle - 120^\circ = 6.81/\angle - 141.8^\circ \text{ A}$$
$$\mathbf{I}_c = \mathbf{I}_a/\angle - 240^\circ = 6.81/\angle - 261.8^\circ \text{ A} = 6.81/\angle 98.2^\circ \text{ A}$$

1.4 Balanced Wye-Delta connection

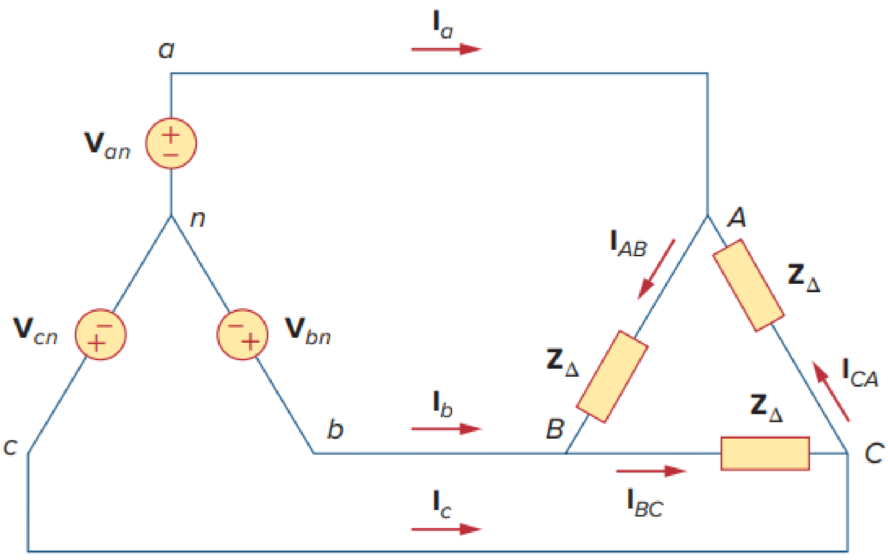


Figure 12.14
Balanced Y-Δ connection.

1. The balanced Y-delta system is shown in Fig. 12.14, where the source is Y-connected and the load is Δ-connected. We assume **positive** sequence.

2. **Source part.** Since the source is **Y-connected**,

   - **Phase voltages:** Positive (abc) sequence assumed.

   $$\mathbf{V}_{an} = V_p\angle 0^\circ, \quad \mathbf{V}_{bn} = V_p\angle -120^\circ, \quad \mathbf{V}_{cn} = V_p\angle -240^\circ = V_p\angle +120^\circ$$

   - **Line voltages:** $\sqrt{3}$ times and leads phase voltage by 30°

   $$\begin{aligned}\mathbf{V}_{ab} &= \sqrt{3}\mathbf{V}_{an}\angle +30^\circ = \sqrt{3}V_p\angle +30^\circ = \mathbf{V}_{AB}\\ \mathbf{V}_{bc} &= \sqrt{3}\mathbf{V}_{bn}\angle +30^\circ = \sqrt{3}V_p\angle -90^\circ = \mathbf{V}_{BC}\\ \mathbf{V}_{ca} &= \sqrt{3}\mathbf{V}_{cn}\angle +30^\circ = \sqrt{3}V_p\angle -210^\circ = \mathbf{V}_{CA}\end{aligned}$$

3. **Load part:** Since the load is Δ-connected, it is straightforward to firstly obtain **phase currents,**

   $$\mathbf{I}_{AB} = \frac{\mathbf{V}_{AB}}{\mathbf{Z}_\Delta}, \quad \mathbf{I}_{BC} = \frac{\mathbf{V}_{BC}}{\mathbf{Z}_\Delta}, \quad \mathbf{I}_{CA} = \frac{\mathbf{V}_{CA}}{\mathbf{Z}_\Delta}$$

   Alternatively, these phase currents can be obtained by applying KVL around loop $aABbna$,

   $$-\mathbf{V}_{an} + \mathbf{Z}_\Delta\mathbf{I}_{AB} + \mathbf{V}_{bn} = 0$$
   $$\mathbf{I}_{AB} = \frac{\mathbf{V}_{an} - \mathbf{V}_{bn}}{\mathbf{Z}_\Delta} = \frac{\mathbf{V}_{ab}}{\mathbf{Z}_\Delta} = \frac{\mathbf{V}_{AB}}{\mathbf{Z}_\Delta}$$

Line currents can be obtained by applying KCL at nodes A, B, and C

$$\mathbf{I}_a = \mathbf{I}_{AB} - \mathbf{I}_{CA}, \quad \mathbf{I}_b = \mathbf{I}_{BC} - \mathbf{I}_{AB}, \quad \mathbf{I}_c = \mathbf{I}_{CA} - \mathbf{I}_{BC}$$

and

$$\begin{aligned}\mathbf{I}_a &= \mathbf{I}_{AB} - \mathbf{I}_{CA} \\ &= \mathbf{I}_{AB}\,(1 - \angle - 240^\circ) = \mathbf{I}_{AB}(1 - \angle + 120^\circ) \\ &= \mathbf{I}_{AB}\left[1 - \left(-\frac{1}{2} + j\frac{\sqrt{3}}{2}\right)\right] = \mathbf{I}_{AB}\sqrt{3}\angle - 30^\circ\end{aligned}$$

Likewise,

$$\begin{aligned}\mathbf{I}_b &= \mathbf{I}_{BC} - \mathbf{I}_{AB} \\ &= \mathbf{I}_{BC}(1 - \angle + 120^\circ) = \mathbf{I}_{BC}\sqrt{3}\angle - 30^\circ \\ \mathbf{I}_c &= \mathbf{I}_{CA} - \mathbf{I}_{BC} \\ &= \mathbf{I}_{CA}(1 - \angle + 120^\circ) = \mathbf{I}_{CA}\sqrt{3}\angle - 30^\circ\end{aligned}$$

To summarize, in Δ-connected load, the line current is $\sqrt{3}$ times larger and lags phase current by 30°,

$$\mathbf{I}_L = \mathbf{I}_p\sqrt{3}\angle - 30^\circ$$

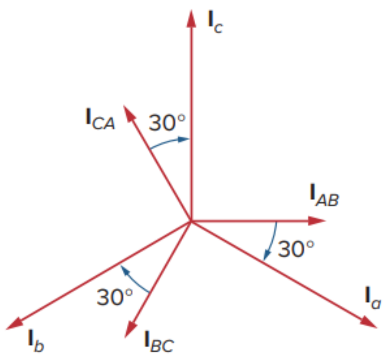


Figure 12.15
Phasor diagram illustrating the relationship between phase and line currents.

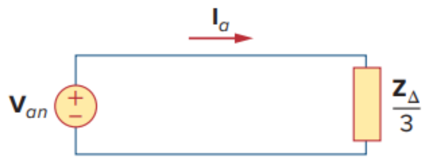


Figure 12.16
A single-phase equivalent circuit of a balanced Y-Δ circuit.

4. Another way to quickly obtain the **line current** is per-phase analysis, which is enabled by transforming the Δ connected load to wye-connected load as in Fig.12.16,

$$\mathbf{Z}_Y = \frac{\mathbf{Z}_\Delta}{3} \quad \rightarrow \quad \mathbf{I}_a = \frac{\mathbf{V}_{an}}{(\mathbf{Z}_\Delta/3)}$$

since

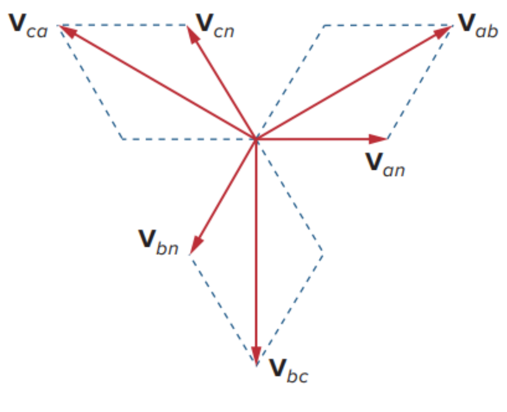
$$\mathbf{I}_a = \frac{\mathbf{V}_{an}}{(\mathbf{Z}_\Delta/3)} = \frac{(\mathbf{V}_{AB}\angle - 30^\circ)/\sqrt{3}}{(\mathbf{Z}_\Delta/3)} = \sqrt{3}\frac{\mathbf{V}_{AB}}{\mathbf{Z}_\Delta}\angle - 30^\circ = \sqrt{3}\mathbf{I}_{AB}\angle - 30^\circ$$

5. The relationship between the phase and line circuit variables are summarized as follows. To memorize this rule,

"Y결선은 전류 같고, 전압 루트3 앞
Δ결선은 전압 같고, 전류 루트3 뒤"

(여기서 '같다'는 line과 phase quantity가 같음을, '앞/뒤'는 phase quantity 기준으로 line quantity의 위상과 크기를 의미함.)

항목	공급 측 (Y결선)	부하 측 (Δ결선)
전압 관계	$V_L = \sqrt{3}V_p\angle 30^\circ$	$V_L = V_\phi$
전류 관계	$I_L = I_\phi$	$I_L = \sqrt{3}I_\phi\angle -30^\circ$



(b)

Figure 12.11
Phasor diagrams illustrating the relationship between line voltages and phase voltages.

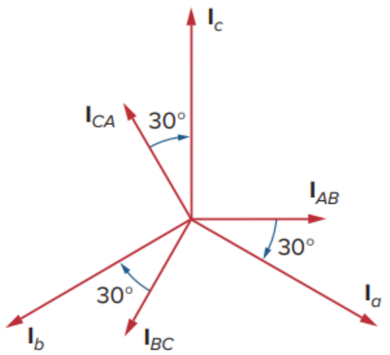


Figure 12.15
Phasor diagram illustrating the relationship between phase and line currents.

6. **Example:** A balanced *abc*-sequence Y-connected source with $\mathbf{V}_{an} = 100\angle 10°$ V is connected to a Δ-connected balanced load $\mathbf{Z}_\Delta = (8+j4)\Omega$ per phase. Calculate the phase and line currents.
Solution:
This can be solved in two ways.

■ **METHOD 1** The load impedance is

$$\mathbf{Z}_\Delta = 8 + j4 = 8.944/\angle 26.57° \ \Omega$$

If the phase voltage $\mathbf{V}_{an} = 100/\angle 10°$, then the line voltage is

$$\mathbf{V}_{ab} = \mathbf{V}_{an}\sqrt{3}/\angle 30° = 100\sqrt{3}/\angle 10° + 30° = \mathbf{V}_{AB}$$

or

$$\mathbf{V}_{AB} = 173.2/\angle 40° \ \text{V}$$

The phase currents are

$$\mathbf{I}_{AB} = \frac{\mathbf{V}_{AB}}{\mathbf{Z}_\Delta} = \frac{173.2/\angle 40°}{8.944/\angle 26.57°} = 19.36/\angle 13.43° \ \text{A}$$

$$\mathbf{I}_{BC} = \mathbf{I}_{AB}/\angle -120° = 19.36/\angle -106.57° \ \text{A}$$
$$\mathbf{I}_{CA} = \mathbf{I}_{AB}/\angle +120° = 19.36/\angle 133.43° \ \text{A}$$

The line currents are

$$\mathbf{I}_a = \mathbf{I}_{AB}\sqrt{3}/\angle -30° = \sqrt{3}(19.36)/\angle(13.43° - 30°) = 33.53/\angle -16.57° \ \text{A}$$

$$\mathbf{I}_b = \mathbf{I}_a/\angle -120° = 33.53/\angle -136.57° \ \text{A}$$
$$\mathbf{I}_c = \mathbf{I}_a/\angle +120° = 33.53/\angle 103.43° \ \text{A}$$

■ **METHOD 2** Alternatively, using single-phase analysis,

$$\mathbf{I}_a = \frac{\mathbf{V}_{an}}{\mathbf{Z}_\Delta/3} = \frac{100/\angle 10°}{2.981/\angle 26.57°} = 33.54/\angle -16.57° \ \text{A}$$

as above. Other line currents are obtained using the *abc* phase sequence.

1.5 Balanced Delta-Delta connection

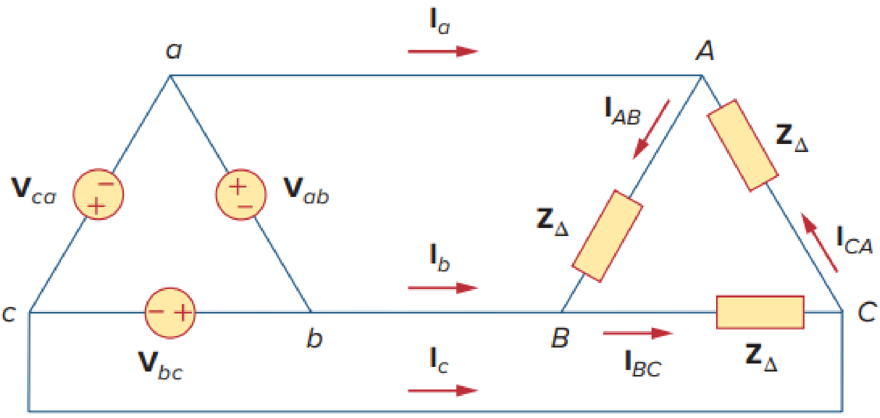


Figure 12.17
A balanced Δ-Δ connection.

1. The source, as well as the load, may be delta-connected, as shown in Fig. 12.17. Our goal is to obtain the phase and line currents, as usual.

2. **Source part.** Since the source is Δ-connected,

   - **Phase voltages:** Positive (abc) sequence assumed.

   $$\mathbf{V}_{ab} = V_p\angle 0^\circ, \quad \mathbf{V}_{bc} = V_p\angle -120^\circ, \quad \mathbf{V}_{ca} = V_p\angle -240^\circ = V_p\angle +120^\circ$$

   - **Line voltages are equal to phase voltages**

   $$\mathbf{V}_{ab} = \mathbf{V}_{AB}, \quad \mathbf{V}_{bc} = \mathbf{V}_{BC}, \quad \mathbf{V}_{ca} = \mathbf{V}_{CA}$$

3. **Load part:** Since the load is Δ-connected, **phase currents** are

   $$\mathbf{I}_{AB} = \frac{\mathbf{V}_{AB}}{\mathbf{Z}_\Delta} = \frac{\mathbf{V}_{ab}}{\mathbf{Z}_\Delta}, \quad \mathbf{I}_{BC} = \frac{\mathbf{V}_{BC}}{\mathbf{Z}_\Delta} = \frac{\mathbf{V}_{bc}}{\mathbf{Z}_\Delta}, \quad \mathbf{I}_{CA} = \frac{\mathbf{V}_{CA}}{\mathbf{Z}_\Delta} = \frac{\mathbf{V}_{ca}}{\mathbf{Z}_\Delta}$$

   Because the load is delta-connected just as in the previous section, some of the formulas derived there apply here. The line currents are obtained from the phase currents by applying KCL at nodes A, B, and C, as we did in the previous section:

   $$\mathbf{I}_a = \mathbf{I}_{AB} - \mathbf{I}_{CA}, \quad \mathbf{I}_b = \mathbf{I}_{BC} - \mathbf{I}_{AB}, \quad \mathbf{I}_c = \mathbf{I}_{CA} - \mathbf{I}_{BC}$$

   Also, as shown in the last section, each line current lags the corresponding phase current by 30°; the magnitude I_L of the line current is $\sqrt{3}$ times the magnitude I_p of the phase current,

   $$\mathbf{I}_L = \sqrt{3}\mathbf{I}_p\angle 30^\circ$$

Example: A balanced Δ-connected load having an impedance $20 - j15\ \Omega$ is connected to a Δ-connected, positive-sequence generator having $V_{ab} = 330\angle 0^\circ$ V. Calculate the phase currents of the load and the line currents.

Solution:
The load impedance per phase is

$$Z_\Delta = 20 - j15 = 25\angle - 36.87^\circ\ \Omega$$

Since $V_{AB} = V_{ab}$, the phase currents are

$$I_{AB} = \frac{V_{AB}}{Z_\Delta} = \frac{330\angle 0^\circ}{25\angle - 36.87^\circ} = 13.2\angle 36.87^\circ\ \text{A}$$

$$I_{BC} = I_{AB}\angle - 120^\circ = 13.2\angle - 83.13^\circ\ \text{A}$$

$$I_{CA} = I_{AB}\angle + 120^\circ = 13.2\angle 156.87^\circ\ \text{A}$$

For a delta load, the line current always lags the corresponding phase current by 30° and has a magnitude $\sqrt{3}$ times that of the phase current. Hence, the line currents are

$$I_a = I_{AB}\sqrt{3}\angle - 30^\circ = (13.2\angle 36.87^\circ)(\sqrt{3}\angle - 30^\circ) = 22.86\angle 6.87^\circ\ \text{A}$$

$$I_b = I_a\angle - 120^\circ = 22.86\angle - 113.13^\circ\ \text{A}$$

$$I_c = I_a\angle + 120^\circ = 22.86\angle 126.87^\circ\ \text{A}$$

1.6 Balanced Delta-Wye Connection

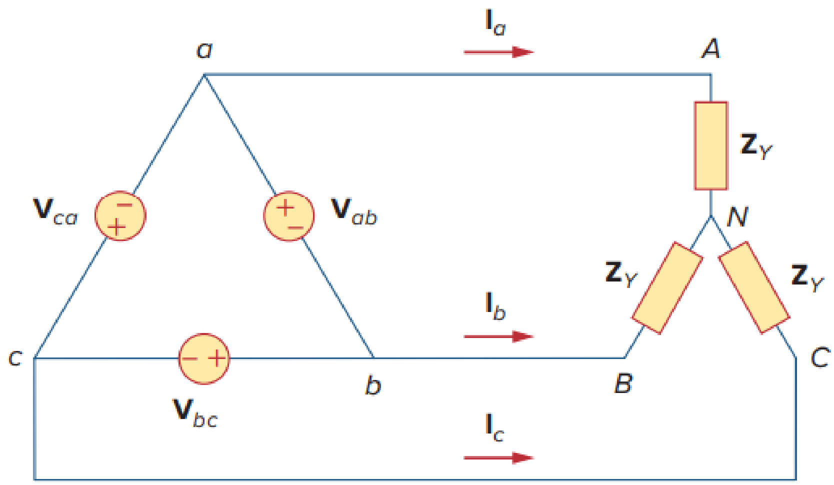


Figure 12.18
A balanced Δ-Y connection.

1. **Source part.** Consider the Δ-Y circuit in Fig. 12.18. Again, assuming the *abc* sequence,
   - **Phase voltages:** Positive (abc) sequence assumed.
   $$\mathbf{V}_{ab} = V_p\angle 0°, \quad \mathbf{V}_{bc} = V_p\angle -120°, \quad \mathbf{V}_{ca} = V_p\angle -240° = V_p\angle +120°$$
   - **Line voltages are equal to phase voltages,**
   $$\mathbf{V}_{ab} = \mathbf{V}_{AB}, \quad \mathbf{V}_{bc} = \mathbf{V}_{BC}, \quad \mathbf{V}_{ca} = \mathbf{V}_{CA}$$
2. **Load part.** We can obtain line currents in three ways.
   - **Method 1.** One way is to apply KVL to loop $aANBba$ in Fig. 12.18, writing
   $$-\mathbf{V}_{ab} + Z_Y\mathbf{I}_a - Z_Y\mathbf{I}_b = 0$$
   $$Z_Y(\mathbf{I}_a - \mathbf{I}_b) = \mathbf{V}_{ab} = \mathbf{V}_p\angle 0°$$
   $$\mathbf{I}_a - \mathbf{I}_b = \frac{\mathbf{V}_p\angle 0°}{Z_Y}$$
   But $\mathbf{I}_b$ lags $\mathbf{I}_a$ by 120°, since we assumed the *abc* sequence; $\mathbf{I}_b = \mathbf{I}_a\angle -120°$. Hence,
   $$\mathbf{I}_a - \mathbf{I}_b = \mathbf{I}_a\left(1 - \frac{1}{\angle -120°}\right)$$
   $$= \mathbf{I}_a\left(1 + \frac{1}{2} + j\frac{\sqrt{3}}{2}\right) = \mathbf{I}_a\sqrt{3}\angle 30°$$
   From this, we obtain the other line currents $\mathbf{I}_b$ and $\mathbf{I}_c$ using the positive phase sequence, i.e., $\mathbf{I}_b = \mathbf{I}_a\angle -120°$, $\mathbf{I}_c = \mathbf{I}_a\angle +120°$. The phase currents are equal to the line currents.

- **Method 2.**

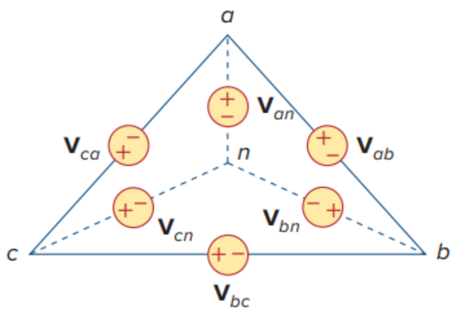


Figure 12.19
Transforming a Δ-connected source to an equivalent Y-connected source.

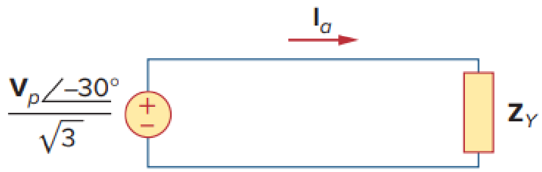


Figure 12.20
The single-phase equivalent circuit.

Another way to obtain the line currents is to replace the deltaconnected source with its equivalent wye-connected source, as shown in Fig. 12.19. We found that the line-to-line voltages of a wye-connected source lead their corresponding phase voltages by 30°. Therefore, we obtain each phase voltage of the equivalent wye-connected source by dividing the corresponding line voltage of the delta-connected source by $\sqrt{3}$ and shifting its phase by $-30°$. Thus, the equivalent wye-connected source has the phase voltages

$$\mathbf{V}_{an} = \frac{V_p}{\sqrt{3}}\angle -30^\circ \quad \mathbf{V}_{bn} = \frac{V_p}{\sqrt{3}}\angle -150^\circ, \quad \mathbf{V}_{cn} = \frac{V_p}{\sqrt{3}}\angle +90^\circ$$

Once the source is transformed to wye, the circuit becomes a wye-wye system. Therefore, we can use the equivalent single-phase circuit shown in Fig. 12.20, from which the line current for phase a is

$$\mathbf{I}_a = \frac{\frac{V_p}{\sqrt{3}}\angle -30^\circ}{Z_Y}$$

- **Method 3.** Alternatively, we may transform the wye-connected load to an equivalent delta-connected load. This results in a delta-delta system. Note that

$$\mathbf{V}_{AN} = \mathbf{I}_a Z_Y = \frac{V_p}{\sqrt{3}}\angle -30^\circ \quad \mathbf{V}_{BN} = \mathbf{V}_{AN}\angle -120^\circ, \quad \mathbf{V}_{CN} = \mathbf{V}_{AN}\angle +120^\circ$$

3. As stated earlier, the delta-connected load is more desirable than the wye-connected load. It is easier to alter the loads in any one phase of the delta-connected loads, as the individual loads are connected directly across the lines.
   However, the delta-connected source is hardly used in practice because any slight imbalance in the phase voltages will result in unwanted circulating currents.

4. **Example:** A balanced Y-connected load with a phase impedance of $40 + j25\ \Omega$ is supplied by a balanced, positive sequence Δ-connected source with a line voltage of 210 V. Calculate the phase currents. Use V_{ab} as a reference.

Solution:
The load impedance is

$$Z_Y = 40 + j25 = 47.17\angle 32^\circ\ \Omega$$

and the source voltage is

$$V_{ab} = 210\angle 0^\circ\ \text{V}$$

When the Δ-connected source is transformed to a Y-connected source,

$$V_{an} = \frac{V_{ab}}{\sqrt{3}}\angle -30^\circ = 121.2\angle -30^\circ\ \text{V}$$

The line currents are

$$I_a = \frac{V_{an}}{Z_Y} = \frac{121.2\angle -30^\circ}{47.12\angle 32^\circ} = 2.57\angle -62^\circ\ \text{A}$$

$$I_b = I_a\angle -120^\circ = 2.57\angle -178^\circ\ \text{A}$$

$$I_c = I_a\angle 120^\circ = 2.57\angle 58^\circ\ \text{A}$$

which are the same as the phase currents.

1.7 Power in a Balanced System

1.7.1 The 1st advantage of three-phase power distribution system.

1. Let us now consider the power in a balanced three-phase system. We begin by examining the instantaneous power absorbed by the load. This requires that the analysis be done in the **time** domain. For a Y-connected load, the phase voltages are

$$v_{AN} = \sqrt{2}V_p \cos\omega t, \quad v_{BN} = \sqrt{2}V_p \cos(\omega t - 120°), \quad v_{CN} = \sqrt{2}V_p \cos(\omega t + 120°)$$

where the factor $\sqrt{2}$ is necessary because V_p has been defined as the **rms value** of the phase voltage. If $Z_Y = Z\angle\theta$, the phase currents lag behind their corresponding phase voltages by θ. Thus,

$$i_a = \sqrt{2}I_p \cos(\omega t - \theta), \quad i_b = \sqrt{2}I_p \cos(\omega t - \theta - 120°), \quad i_c = \sqrt{2}I_p \cos(\omega t - \theta + 120°)$$

where I_p is the **rms value** of the phase current.

2. The total instantaneous power in the load is the sum of the instantaneous powers in the three phases; that is,

$$\begin{aligned} p = p_a + p_b + p_c = v_{AN}i_a &+ v_{BN}i_b + v_{CN}i_c \\ &= 2V_pI_p\,[\,\cos\omega t\cos(\omega t - \theta) + \\ &\qquad \cos(\omega t - 120°)\cos(\omega t - \theta - 120°) + \\ &\qquad \cos(\omega t + 120°)\cos(\omega t - \theta + 120°) \end{aligned}$$

Applying the trigonometric identity

$$\cos A \cos B = \frac{1}{2}[\cos(A+B) + \cos(A-B)]$$

gives

$$\begin{aligned} p &= V_pI_p\,[3\cos\theta + \cos(2\omega t - \theta) + \cos(2\omega t - \theta - 240°) + \cos(2\omega t - \theta + 240°)] \\ &= V_pI_p\,[3\cos\theta + \cos\alpha + \cos\alpha\cos 240° + \sin\alpha\sin 240° + \cos\alpha\cos 240° - \sin\alpha\sin 240°] \\ &\text{where } \alpha = 2\omega t - \theta \\ &= V_pI_p\left[3\cos\theta + \cos\alpha + 2\left(-\frac{1}{2}\right)\cos\alpha\right] \\ &= 3V_pI_p\cos\theta \end{aligned}$$

3. **The first advantage of the three-phase power system**
Thus the total instantaneous power in a balanced three-phase system is **constant—it does not change with time as the instantaneous power of each phase does**. This result is true whether the load is Y- or Δ-connected. This is one important reason for using a three-phase system to generate and distribute power.

4. **Per-phase power.** Since the total instantaneous power is independent of time, the average power per phase P_p for either the Δ-connected load or the Y-connected load is $p/3$, or

$$P_p = V_p I_p \cos\theta$$

and the reactive power per phase is

$$Q_p = V_p I_p \sin\theta$$

The apparent power per phase is

$$S_p = V_p I_p$$

The complex power per phase is

$$S_p = P_p + jQ_p = V_p I_p^*$$

where V_p and I_p are the phase voltage and phase current with magnitudes V_p and I_p, respectively.

5. **The total average power** is the sum of the average powers in the phases:

$$P = P_a + P_b + P_c = 3P_p = 3V_p I_p \cos\theta = \sqrt{3} V_L I_L \cos\theta$$

For a Y-connected load, $I_L = I_p$ but $V_L = \sqrt{3} V_p$, whereas for a Δ-connected load, $I_L = \sqrt{3} I_p$ but $V_L = V_p$. Thus, the equation above applies for both Y-connected and Δ-connected loads. Similarly, the total reactive power is

$$Q = 3V_p I_p \sin\theta = 3Q_p = \sqrt{3} V_L I_L \sin\theta$$

and the total complex power is

$$S = 3S_p = 3V_p I_p^* = 3I_p^2 Z_p = \frac{3V_p^2}{Z_p^*}$$

where $Z_p = Z_p/\theta$ is the load impedance per phase. (Z_p could be Z_Y or Z_Δ.) Alternatively,

$$S = P + jQ = \sqrt{3} V_L I_L \angle\theta$$

Remember that V_p, I_p, V_L, and I_L are all rms values and that θ is the angle of the load impedance or the angle between the phase voltage and the phase current.

Example: Power Factor Correction with three phase loads.
Two balanced loads are connected to a 240-kV rms 60-Hz line, as shown in Fig. 12.22(a).

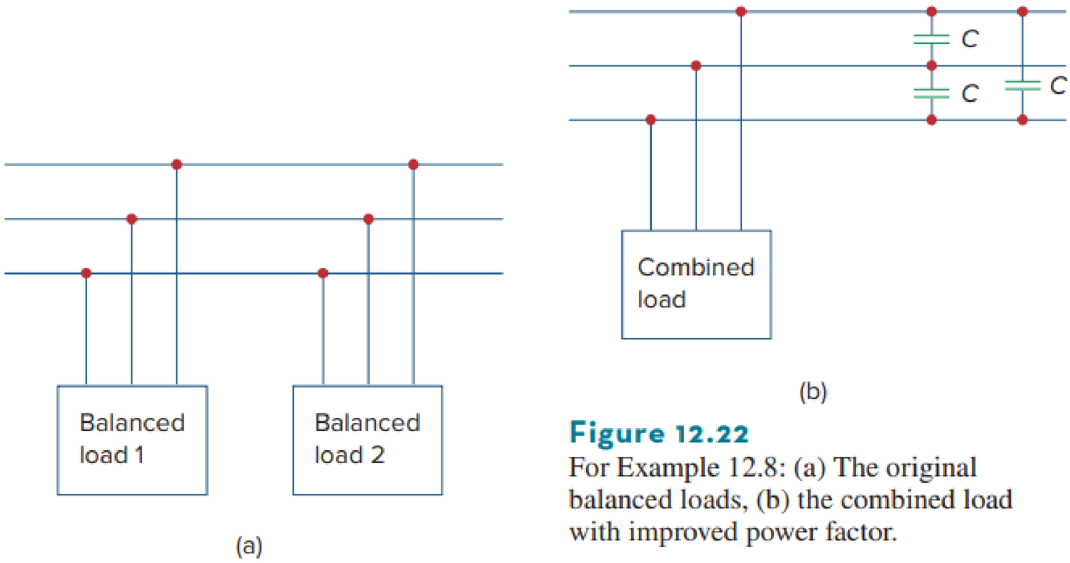


Figure 12.22
For Example 12.8: (a) The original balanced loads, (b) the combined load with improved power factor.

Load 1 draws 30 kW at a power factor of 0.6 lagging, while load 2 draws 45 kVAR at a power factor of 0.8 lagging. Assuming the *abc* sequence, determine:
(a) the complex, real, and reactive powers absorbed by the combined load,
(b) the line currents, and
(c) the kVAR rating of the three capacitors Δ-connected in parallel with the load that will raise the power factor to 0.9 lagging and the capacitance of each capacitor.

Solution:

(a) For load 1, given that $P_1 = 30$ kW and $\cos\theta_1 = 0.6$, then $\sin\theta_1 = 0.8$. Hence,

$$S_1 = \frac{P_1}{\cos\theta_1} = \frac{30 \text{ kW}}{0.6} = 50 \text{ kVA}$$

and $Q_1 = S_1 \sin\theta_1 = 50(0.8) = 40$ kVAR. Thus, the complex power due to load 1 is

$$\boxed{S_1 = P_1 + jQ_1 = 30 + j40 \text{ kVA}} \tag{12.8.1}$$

For load 2, if $Q_2 = 45$ kVAR and $\cos\theta_2 = 0.8$, then $\sin\theta_2 = 0.6$. We find

$$S_2 = \frac{Q_2}{\sin\theta_2} = \frac{45 \text{ kVAR}}{0.6} = 75 \text{ kVA}$$

and $P_2 = S_2 \cos\theta_2 = 75(0.8) = 60$ kW. Therefore the complex power due to load 2 is

$$\boxed{S_2 = P_2 + jQ_2 = 60 + j45 \text{ kVA}} \tag{12.8.2}$$

From Eqs. (12.8.1) and (12.8.2), the total complex power absorbed by the load is

$$\boxed{S = S_1 + S_2 = 90 + j85 \text{ kVA} = 123.8\angle 43.36^\circ \text{ kVA}} \tag{12.8.3}$$

which has a power factor of $\cos 43.36^\circ = 0.727$ lagging. The real power is then 90 kW, while the reactive power is 85 kVAR.

(b) Since $S = 3\left(\frac{240 \text{ kV}}{\sqrt{3}}\right) I_L = \sqrt{3}(240 \text{ kV}) I_L$,

$$\boxed{I_L = \frac{S}{\sqrt{3}(240,000)}} \tag{12.8.4}$$

We apply this to each load keeping in mind that the magnitude of the phase voltages is equal to $(240/\sqrt{3})$ kV. For load 1,

$$I_{L1} = \frac{50,000}{\sqrt{3} \cdot 240,000} = 120.28 \text{ mA}$$

Since the power factor is lagging, the line current lags the line voltage by $\theta_1 = \cos^{-1} 0.6 = 53.13^\circ$. Thus,

$$\mathbf{I}_{a1} = 120.28\angle -53.13^\circ$$

For load 2,

$$I_{L2} = \frac{75,000}{\sqrt{3} \cdot 240,000} = 180.42 \text{ mA}$$

and the line current lags the line voltage by $\theta_2 = \cos^{-1} 0.8 = 36.87^\circ$. Hence,

$$\mathbf{I}_{a2} = 180.42\angle -36.87^\circ$$

The total line current is

$$\mathbf{I}_a = \mathbf{I}_{a1} + \mathbf{I}_{a2} = 120.28\angle -53.13^\circ + 180.42\angle -36.87^\circ$$

$$= (72.168 - j96.224) + (144.336 - j108.252)$$

$$= 216.5 - j204.472 = \boxed{297.8\angle - 43.36^\circ \text{ mA}}$$

Alternatively, we could obtain the current from the total complex power using Eq. (12.8.4) as

$$I_L = \frac{123,800}{\sqrt{3} \cdot 240,000} = 297.82 \text{ mA}$$

and

$$\mathbf{I}_a = 297.82\angle - 43.36^\circ \text{ mA}$$

which is the same as before. The other line currents, $\mathbf{I}_b$ and $\mathbf{I}_c$, can be obtained according to the *abc* sequence (i.e., $\mathbf{I}_b = 297.82\angle{-163.36^\circ}$ mA and $\mathbf{I}_c = 297.82\angle 76.64^\circ$ mA).

(c) We can find the reactive power needed to bring the power factor to 0.9 lagging using Eq. (11.59),

$$Q_C = P(\tan\theta_{\text{old}} - \tan\theta_{\text{new}})$$

where $P = 90$ kW, $\theta_{\text{old}} = 43.36^\circ$, and $\theta_{\text{new}} = \cos^{-1} 0.9 = 25.84^\circ$. Hence,

$$Q_C = 90,000(\tan 43.36^\circ - \tan 25.84^\circ) = 41.4 \text{ kVAR}$$

This reactive power is for the three capacitors. For each capacitor, the rating Q'_C is 13.8 kVAR. From Eq. (11.60), the required capacitance is

$$C = \frac{Q'_C}{\omega V_{\text{rms}}^2}$$

Since the capacitors are Δ-connected as shown in Fig. 12.22(b), V_{rms} in the above formula is the line-to-line or line voltage, which is 240 kV. Thus,

$$\boxed{C = \frac{13,800}{(2\pi 60)(240,000)^2} = 635.5 \text{ pF}}$$

1.7.2 The 2nd advantage of three-phase power distribution system.

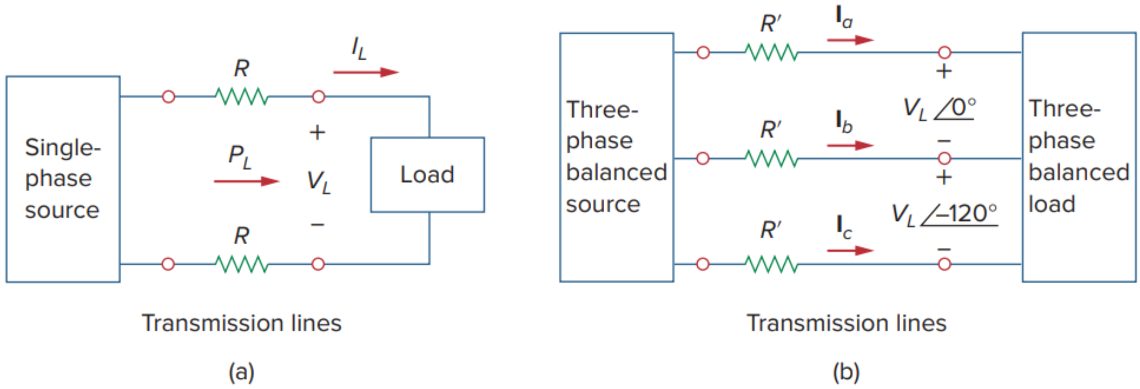


Figure 12.21
Comparing the power loss in (a) a single-phase system, and (b) a three-phase system.

1. A second major advantage of three-phase systems for power distribution is that the three-phase system uses a **lesser amount of wire** than the single-phase system for the **same line voltage** V_L **and the same absorbed power** P_L.

2. We will compare these cases and assume in both that the wires are of the same material (e.g., copper with resistivity ρ), of the same length ℓ, and that the loads are resistive (i.e., unity power factor). For the two-wire single-phase system in Fig. 12.21(a), $I_L = P_L/V_L$, so the power loss in the two wires is

$$P_{\text{loss}} = 2I_L^2 R = 2R\frac{P_L^2}{V_L^2}$$

For the three-wire three-phase system in Fig. 12.21(b), $I'_L = |I_a| = |I_b| = |I_c| = P_L/\sqrt{3}V_L$. The power loss in the three wires is

$$P'_{\text{loss}} = 3(I'_L)^2 R' = 3R'\frac{P_L^2}{3V_L^2} = R'\frac{P_L^2}{V_L^2}$$

For the same total power delivered P_L and same line voltage V_L,

$$\frac{P_{\text{loss}}}{P'_{\text{loss}}} = \frac{2R}{R'}$$

3. But $R = \rho\ell/\pi r^2$ and $R' = \rho\ell/\pi r'^2$, where r and r' are the radii of the wires. Thus,

$$\frac{P_{\text{loss}}}{P'_{\text{loss}}} = \frac{2r'^2}{r^2}$$

If the same power loss is tolerated in both systems, then $r^2 = 2r'^2$. The ratio of material required is determined by the number of wires and their volumes, so

$$\frac{\text{Material for single-phase}}{\text{Material for three-phase}} = \frac{2(\pi r^2 \ell)}{3(\pi r'^2 \ell)} = \frac{2r^2}{3r'^2} = \frac{2}{3}(2) = 1.333$$

since $r^2 = 2r'^2$. This shows that the single-phase system uses 33 percent more material than the three-phase system or that the three-phase system uses only 75 percent of the material used in the equivalent single-phase system. In other words, considerably less material is needed to deliver the same power with a three-phase system than is required for a single-phase system.

Chapter 2

Unbalanced Three phase circuits

2.1 Balanced sources with unbalanced loads

1. An unbalanced system is caused by two possible situations:
   (1) The source voltages are not equal in magnitude and/or differ in phase by angles that are unequal, or
   (2) load impedances are unequal.
   To begin with, we will assume balanced source voltages, but an unbalanced load.

2. Unbalanced three-phase systems are solved by direct application of mesh and nodal analysis. Figure 12.23 shows an example of an unbalanced three-phase system that consists of balanced source voltages (not shown in the figure) and an unbalanced Y-connected load (shown in the figure).

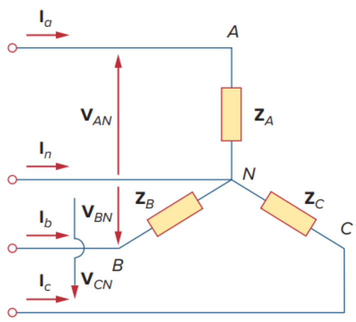


Figure 12.23
Unbalanced three-phase Y-connected load.

   Since the load is unbalanced, $\mathbf{Z}_A$, $\mathbf{Z}_B$, and $\mathbf{Z}_C$ are not equal. The line currents are determined by Ohm's law as

$$\mathbf{I}_a = \frac{\mathbf{V}_{AN}}{Z_A}, \quad \mathbf{I}_b = \frac{\mathbf{V}_{BN}}{Z_B}, \quad \mathbf{I}_c = \frac{\mathbf{V}_{CN}}{Z_C}$$

3. This set of unbalanced line currents produces current in the **neutral line**, which is not zero as in a balanced system. Applying KCL at node N gives the neutral line current as

$$\mathbf{I}_n = -(\mathbf{I}_a + \mathbf{I}_b + \mathbf{I}_c)$$

4. In a three-wire system where the neutral line is absent, we can still find the line currents $\mathbf{I}_a$, $\mathbf{I}_b$, and $\mathbf{I}_c$ using mesh analysis. At node N, KCL must be satisfied so that $\mathbf{I}_a+\mathbf{I}_b+\mathbf{I}_c = 0$ in this case. The same could be done for an unbalanced Δ-Y, Y-Δ, or Δ-Δ three-wire system. As mentioned earlier, in long distance power transmission, conductors in multiples of three (multiple three-wire systems) are used, with the earth itself acting as the neutral conductor.

5. To calculate power in an unbalanced three-phase system requires that we find the power in each phase using Eqs. (12.46) to (12.49). The total power is not simply three times the power in one phase but the sum of the powers in the three phases.

6. **Example: Finding the neutral current due to unbalanced loads.**
The unbalanced Y-load of Fig. 12.23 has balanced voltages of 100 V and the *acb* sequence. Calculate the line currents and the neutral current. Take $Z_A = 15\ \Omega$, $Z_B = 10 + j5\ \Omega$, $Z_C = 6 - j8\ \Omega$.

Solution:
Using Eq. (12.59), the line currents are

$$\mathbf{I}_a = \frac{100\angle 0^\circ}{15} = 6.67\angle 0^\circ \text{ A}$$

$$\mathbf{I}_b = \frac{100\angle 120^\circ}{10 + j5} = \frac{100\angle 120^\circ}{11.18\angle 26.56^\circ} = 8.94\angle 93.44^\circ \text{ A}$$

$$\mathbf{I}_c = \frac{100\angle -120^\circ}{6 - j8} = \frac{100\angle -120^\circ}{10\angle -53.13^\circ} = 10\angle -66.87^\circ \text{ A}$$

Using Eq. (12.60), the current in the neutral line is

$$\mathbf{I}_n = -(\mathbf{I}_a + \mathbf{I}_b + \mathbf{I}_c) = -(6.67 - 0.54 + j8.92 + 3.93 - j9.2)$$

$$= -10.06 + j0.28 = 10.06\angle 178.4^\circ \text{ A}$$

7. **Example:** For the unbalanced circuit in Fig. 12.25, find: (a) the line currents, (b) the total complex power absorbed by the load, and (c) the total complex power absorbed by the source.

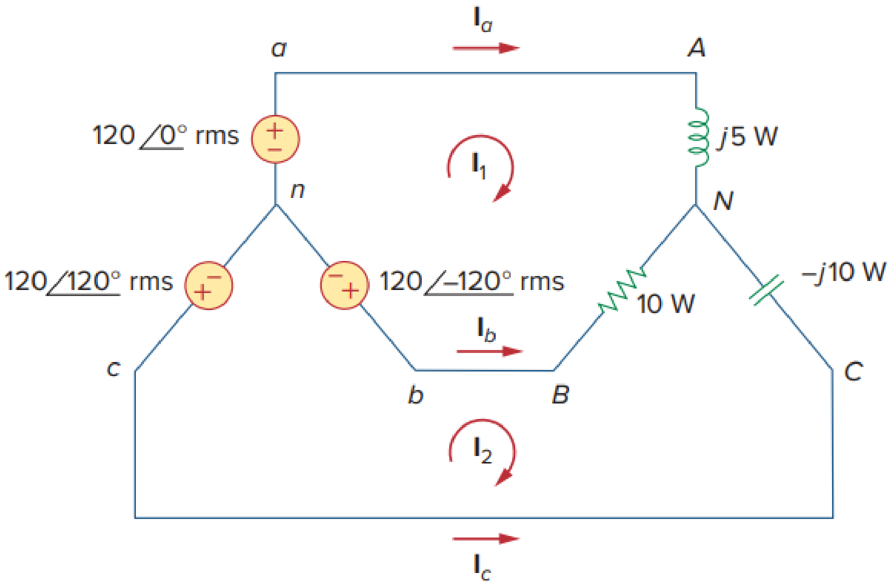


Figure 12.25
For Example 12.10.

Solution:
(a) We use mesh analysis to find the required currents. For mesh 1,

$$\underline{120\angle -120^\circ - 120\angle 0^\circ + (10+j5)\mathbf{I}_1 - 10\mathbf{I}_2} = 0$$

or

$$(10+j5)\mathbf{I}_1 - 10\mathbf{I}_2 = 120\sqrt{3}\angle 30^\circ \tag{12.10.1}$$

For mesh 2,

$$\underline{120\angle 120^\circ - 120\angle -120^\circ + (10-j10)\mathbf{I}_2 - 10\mathbf{I}_1} = 0$$

or

$$-10\mathbf{I}_1 + (10-j10)\mathbf{I}_2 = 120\sqrt{3}\angle -90^\circ \tag{12.10.2}$$

Equations (12.10.1) and (12.10.2) form a matrix equation:

$$\begin{bmatrix} 10+j5 & -10 \\ -10 & 10-j10 \end{bmatrix} \begin{bmatrix} \mathbf{I}_1 \\ \mathbf{I}_2 \end{bmatrix} = \begin{bmatrix} 120\sqrt{3}\angle 30^\circ \\ 120\sqrt{3}\angle -90^\circ \end{bmatrix}$$

The determinants are

$$\Delta = \begin{vmatrix} 10+j5 & -10 \\ -10 & 10-j10 \end{vmatrix} = 50 - j50 = 70.71\angle -45^\circ$$

$$\Delta_1 = \begin{vmatrix} 120\sqrt{3}\angle 30^\circ & -10 \\ 120\sqrt{3}\angle -90^\circ & 10-j10 \end{vmatrix} = 207.85(13.66 - j13.66) = 4015.23\angle -45^\circ$$

$$\Delta_2 = \begin{vmatrix} 10+j5 & 120\sqrt{3}\angle 30^\circ \\ -10 & 120\sqrt{3}\angle -90^\circ \end{vmatrix} = 207.85(13.66 - j5) = 3023.4\angle -20.1^\circ$$

The mesh currents are

$$\mathbf{I}_1 = \frac{\Delta_1}{\Delta} = \frac{4015.23\angle -45^\circ}{70.71\angle -45^\circ} = 56.78 \text{ A}$$

$$\mathbf{I}_2 = \frac{\Delta_2}{\Delta} = \frac{3023.4\angle -20.1^\circ}{70.71\angle -45^\circ} = 42.75\angle 24.9^\circ \text{ A}$$

The line currents are

$$\mathbf{I}_a = \mathbf{I}_1 = 56.78 \text{ A}, \quad \mathbf{I}_c = -\mathbf{I}_2 = 42.75\angle -155.1^\circ \text{ A}$$

$$\mathbf{I}_b = \mathbf{I}_2 - \mathbf{I}_1 = 38.78 + j18 - 56.78 = 25.46\angle 135^\circ \text{ A}$$

(b) We can now calculate the complex power absorbed by the load. For phase A,

$$S_A = \mathbf{I}_a^2 Z_A = (56.78)^2 (j5) = j16,120 \text{ VA}$$

For phase B,

$$S_B = \mathbf{I}_b^2 Z_B = (25.46)^2 (10) = 6480 \text{ VA}$$

For phase C,

$$S_C = \mathbf{I}_c^2 Z_C = (42.75)^2(-j10) = -j18,276 \text{ VA}$$

The total complex power absorbed by the load is

$$S_L = S_A + S_B + S_C = 6480 - j2156 \text{ VA}$$

(c) We check the result above by finding the power absorbed by the source. For the voltage source in phase a,

$$S_a = -\mathbf{V}_{an}\mathbf{I}_a^* = -(120\angle 0^\circ)(56.78) = -6813.6 \text{ VA}$$

For the source in phase b,

$$S_b = -\mathbf{V}_{bn}\mathbf{I}_b^* = -(120\angle{-120^\circ})(25.46\angle{-135^\circ}) = -3055.2\angle 105^\circ = 790 - j2951.1 \text{ VA}$$

For the source in phase c,

$$S_c = -\mathbf{V}_{cn}\mathbf{I}_c^* = -(120\angle 120^\circ)(42.75\angle 155.1^\circ) = -5130\angle 275.1^\circ = -4560.03 + j5109.7 \text{ VA}$$

The total complex power absorbed by the three-phase source is

$$S_s = S_a + S_b + S_c = -6480 + j2156 \text{ VA}$$

showing that $S_s + S_L = 0$ and confirming the conservation principle of ac power.

Example: Find the line currents in the unbalanced three-phase circuit of Fig. 12.26 and the real power absorbed by the load.

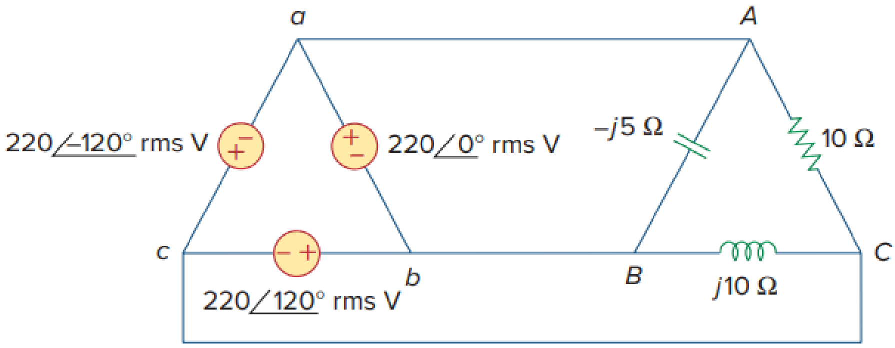


Figure 12.26
For Practice Prob. 12.10.

Answer: $64\angle 80.1^\circ$ A, $38.1\angle -60^\circ$ A, $42.5\angle -135^\circ$ A, 4.84 kW.